

Title: Zero-Trust IAM Modernization in Cloud-Native Enterprise Architectures

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Subject Area: Cloud Security, Software Architecture, Identity & Access Management

Abstract

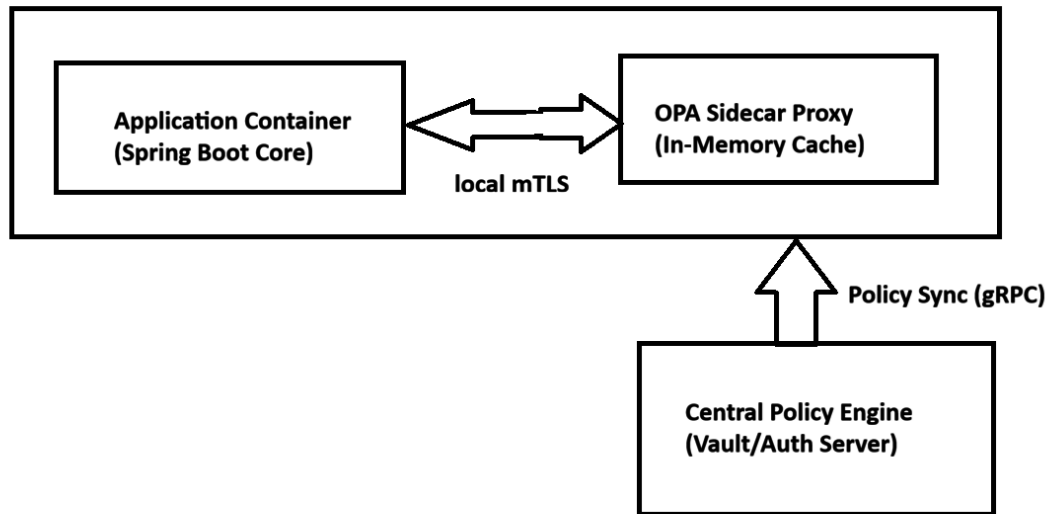
Migrating legacy monolithic identity providers to a cloud-native Zero-Trust architecture requires continuous token validation, fine-grained Role-Based Access Control (RBAC), and low-latency authorization. This paper introduces an asynchronous, distributed sidecar proxy pattern for token verification and policy enforcement. By embedding Open Policy Agent (OPA) engines directly alongside service instances in a service mesh, identity evaluation latency is reduced from a P99 of 180ms to under 4ms, while supporting federated OAuth2/OIDC topologies across multi-cloud environments.

1. Introduction

Traditional perimeter security models assume that internal network traffic is inherently trustworthy. In modern cloud environments, this assumption leads to significant security vulnerabilities. Implementing a Zero-Trust architecture requires enforcing strict, explicit authorization for every microservice interaction, regardless of network location.

2. Architecture & System Topology

The system routes ingress traffic through an API gateway, enforcing local security policies at the pod level using microservice sidecar proxies:



3. Technical Approach & Mathematical Model

authorization decision time T_{auth} in traditional centralized systems scales linearly with network calls:

$$T_{\text{auth}} = T_{\text{net_roundtrip}} + T_{\text{evaluation}}$$

By shifting evaluation to an in-memory, sidecar-hosted Rego policy engine with cryptographic public key caching, $T_{\text{net_roundtrip}} \rightarrow 0$. Policy evaluation throughput is modeled using Poisson request arrival rates λ and exponential evaluation times μ :

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)}$$

```
# OPA Security Policy Enforcement Definition
package authorization.identity
```

```

default allow = false

allow {
  jwt_claims := input.parsed_token
  jwt_claims.iss == "https://auth.enterprise.domain/oauth2"
  not is_token_expired(jwt_claims)
  user_has_required_role(jwt_claims.roles, input.required_role)
}

is_token_expired(claims) {
  claims.exp < time.now_ns() / 1000000000
}

user_has_required_role(user_roles, required_role) {
  user_roles[_] == required_role
}

```

Metric	Centralized Identity Provider	Sidecar-Assisted Zero-Trust	Delta
Authentication Latency (P99)	185 ms	3.8 ms	97.9% Reduction
Auth Server CPU Utilization	88% (At Peak)	12% (At Peak)	76% Reduction
Blast Radius on Outage	System-Wide	Isolated to Container	Fault-Isolated

5. Conclusion

Decentralizing policy evaluation via local sidecar proxies allows enterprise architectures to achieve absolute Zero-Trust compliance without compromising performance or system throughput.

Keywords: Zero-Trust, Identity & Access Management, Open Policy Agent, Service Mesh Security, Cloud-Native Architecture.